

Benjamin Mellin

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PROFESSIONAL SUMMARY

Materials Science and Engineering student at the University of Florida with direct experience in hypersonic simulation, thermal-structural analysis, composite fabrication, and metallurgical processing. Led materials selection and prototype development on a Mach 20 atmospheric re-entry project in partnership with Honeywell Aerospace. Applied computational and experimental methods to complex aerospace material systems and communicated findings directly to industry sponsors.

EDUCATION

University of Florida

Bachelor of Science in Materials Science and Engineering. Cum Laude.

Gainesville, FL

May 2026

University of Florida

Associate of Arts.

Gainesville, FL

Dec. 2023

EXPERIENCE

Lead Materials Engineer

Aug. 2025 – May 2026

Honeywell Aerospace | Atmospheric Re-Entry Control

Clearwater, FL

- Led the physical prototype sub-team and owned the full materials selection workstream within a 7-person team, developing a custom S.M.A.R.T. (Simulation and Machine-learning Assisted Redesign and Testing) methodology that replaced traditional trial-and-error with an automated, data-driven design loop using Kriging algorithms and hypersonic simulation to optimize a magnesium chassis for atmospheric demise.
- Engineered deployable flaps to induce intentional shock-shock interference during Mach 20 descent, achieving a design validated to capture 64.5 MW of heat and spike structural temperatures to 1,555 K, demonstrating full demisability compliance in partnership with Honeywell Aerospace.
- Documented and communicated team progress through a weekly public blog, maintaining technical transparency with Honeywell Aerospace industry sponsors throughout the design cycle.

Team Lead

Mar. 2025 – Apr. 2025

1075 Steel Heat Treatment Design

Gainesville, FL

- Directed a full heat treatment design process for 1075 steel, including literature review, phase diagram analysis, and TTT diagram interpretation to select an optimal heat-and-quench-plus-temper processing route.
- Martensitically transformed and tempered the steel to achieve target hardness; presented results as a research poster.

Lead Composite Engineer

Jan. 2025 – Jan. 2025

Fiberglass Lay-Up

Gainesville, FL

- Designed and fabricated fiberglass sheet and tube constructions using wet lay-up, authoring standard operating procedures for each production step.
- Validated composite performance via Charpy impact, tensile (Instron), and 3-point bending tests, achieving results within 80% of theoretical maximum; documented findings in a formal technical memo.

Team Lead

Aug. 2024 – Dec. 2024

Engineering Innovation

Gainesville, FL

- Led a cross-functional team through the full product development lifecycle, from need analysis and market research through brainstorming, prototyping, and investor-style pitch.
- Conceptualized and pitched MindMate, a wearable AI assistant for elderly task management; developed a comprehensive innovation playbook and presented to a faculty jury in the UF innovation department.

Volunteer Foster

May 2025 – Present

Low Rider Rescue

Wesley Chapel, FL

- Provide structured foster care for dogs, tracking health and behavioral progress and communicating observations to rescue coordinators and prospective adopters.

SKILLS

Software & Data Tools: ANSYS, SolidWorks, MATLAB, Python, GRANTA, Excel

Engineering: Thermal-Structural Analysis, Material Characterization, Failure Analysis, Composite Production, Product Development, DLP & FDM 3D Printing, Arduino, Deconstruction

Testing Techniques: Tensile (Instron UTS), Charpy Impact, 3-Point Bending, Scanning Electron Microscopy, X-ray Diffraction, Fourier Transform Infrared (FTIR) Spectroscopy, Degradation Testing

Communication: Technical Writing (SOPs, research papers, failure analysis reports), Public Speaking, Poster and PowerPoint Presentation, Group Leadership, Literature Review, Six Sigma Quality White Belt